The Representation Bottleneck: A Perturbation Audit of Module-Graph Features for Genomic Ecosystem Comparison

Ajax Davis

April 2026

ABSTRACT

We present the results of a systematic perturbation audit of the module-graph representation that has served as the computational substrate for ten versions of Spectral Ecology and three prior symbiogenesis algorithms. This is a **negative result paper**. The audit applies three complementary stress tests—a coverage ladder (subsampling at 100%, 80%, 60%, and 40% of 50,000 reads), cross-sample variance analysis, and a read-order shuffle test—to the 15 features extracted from k-mer co-occurrence graphs constructed for six members of the CEPH 1463 pedigree. The findings are unambiguous and sobering. First, only 5 of 15 features achieve self-perturbation stability (coefficient of variation below 0.15 across coverage levels): module count, edge density, mean degree, mean curvature, and median curvature. The remaining 10 features, including all five ecological role counts that several algorithm versions weighted heavily, are *unstable under subsampling within the same sample*. Second, the signal-to-nuisance ratio—defined as cross-sample variance divided by self-perturbation variance—exceeds 3 for **zero** of 15 features. The best ratio is 2.93 (edge density); the worst is 0.59 (scaffold role count). Four features, including two of the three curvature statistics that Spectral Ecology v9 scored, have ratios below 1.0, meaning that variation introduced by subsampling the *same* genome exceeds variation between *different* genomes. For these features, the representation is literally noisier than the signal it is supposed to carry. Third, and most devastatingly, shuffling read order—a transformation that changes zero biological content—produces catastrophic changes in graph structure: module count increases by 45% to 184%, triangle count increases by 2% to 607%, and edge count increases by 27% to 264%. The module-graph representation is not merely noisy; it is *non-deterministic* with respect to read ordering, a property that no downstream invariant can correct. These results explain why the ten-version Spectral Ecology arc, despite producing genuine theoretical advances in coherence fields, intrinsic curvature distributions, holonomy deficits, and Sinkhorn-regularized optimal transport, never achieved a positive weakest-pair gap: the scoring algorithms were optimizing invariants on an unstable representation. The downstream sophistication was outrunning upstream identifiability. We provide detailed tables of all 15 features across all three tests, analyze the two "least-bad" features (edge density at ratio 2.93 and median curvature at ratio 2.47) as potential starting points for a stabilized representation, and offer concrete recommendations for the next phase: deterministic module extraction, coverage-matched subsampling, rank-based normalization, and an acceptance criterion requiring signal-to-nuisance ratio above 3 for every scored feature before any new algorithm version is attempted. This paper changes the direction of the yalumba research program. It is, we argue, the most important experiment the project has conducted.

Contents

1 Introduction **6**
1.1 Why Negative Results Matter . . . 6
1.2 The Ten-Version Arc as Context . . . 6
1.3 The Feedback That Prompted This Audit . . . 7
1.4 Why This Is the Most Important Experiment . . . 8
1.5 Relationship to Passing Algorithms . . . 8

2 Background **9**
2.1 The Module-Graph Representation . . . 9
2.2 What Ten Versions Assumed . . . 10
2.3 The Concept of Signal-to-Nuisance Ratio . . . 10
2.4 Why Perturbation Testing Matters . . . 11

3 Methods **12**
3.1 Dataset . . . 12
3.2 Coverage Ladder . . . 12
3.3 Shuffle Test . . . 13
3.4 Feature Extraction . . . 13
3.5 Self-Perturbation CV . . . 13
3.6 Cross-Sample CV . . . 14
3.7 Signal-to-Nuisance Ratio . . . 14
3.8 Experimental Protocol . . . 14
3.9 Stability Classification . . . 14
3.10 Verdict Classification for SNR . . . 15

4 Results **15**
4.1 Self-Perturbation Analysis . . . 15
4.2 Cross-Sample Variance . . . 16
4.3 Signal-to-Nuisance Ratio . . . 17
4.4 The Curvature Shock . . . 19
4.5 Shuffle Test Results . . . 19
4.6 Per-Sample Shuffle Detail . . . 20
4.7 What the Shuffle Test Proves . . . 21
4.8 The Two Least-Bad Features . . . 21
4.8.1 Edge Density (SNR = 2.93) . . . 21

4.8.2 Curvature Median (curvP50, SNR = 2.47) 22
4.9 Summary of Results 22

5 Discussion **22**
5.1 What the Ten-Version Arc Was Actually Optimizing 22
5.2 Why the Spouse Confound Persists 24
5.3 The Shuffle Catastrophe 24
5.3.1 Magnitude of the Effect 25
5.4 Why Role Assignment Is Almost Entirely Noise 25
5.5 The Coverage Ladder: What Subsampling Reveals 26
5.6 Salvageable Features 27
5.7 What Canonicalization Would Need to Achieve 27
5.8 Connection to the Theoretical Framework 28
5.9 Implications for Other Structural Genomics Approaches 29
5.10 This Is a Strong Scientific Result, Not a Failure 30
5.11 Limitations of This Audit 30

6 Recommendations for the Next Phase **31**
6.1 Representation Canonicalization 31
6.2 Coverage-Matched Subsampling 32
6.3 Rank-Based or Quantile-Normalized Features 32
6.4 Threshold Sweeps 32
6.5 The Acceptance Criterion for v11 33
6.6 Feature Design Principles 33
6.7 A Proposed Verification Pipeline 34

7 Conclusion **34**

1 Introduction

1.1 Why Negative Results Matter

The history of computational science is littered with methods that optimized the wrong objective on the wrong substrate. In machine learning, feature engineering on unstable representations produced years of incremental papers before practitioners learned to audit the representation itself [1]. In genomics, early reference-free methods achieved impressive-looking metrics on benchmarks whose variance properties were never characterized [10]. In ecology, decades of community comparison used indices whose sensitivity to sampling effort exceeded their sensitivity to actual community differences [2, 17].

The most important experiment in a research program is not the one that succeeds. It is the one that reveals what success was being measured against.

This paper reports such an experiment. After ten versions of Spectral Ecology—a research arc spanning sparse graphs, dense co-occurrence graphs, optimal transport, coherence fields, persistence fields, Sinkhorn regularization, holonomy deficits, intrinsic curvature, and hybrid invariants—we step back and ask a question that should have been asked before version 1:

▶ **Key Finding**

Is the module-graph representation stable enough for *any* downstream invariant to distinguish biological signal from representation noise?

The answer is no.

1.2 The Ten-Version Arc as Context

Table 1 summarizes the Spectral Ecology program. Each version introduced increasingly sophisticated mathematical machinery: Laplacian eigendecomposition (v1–v2), Wasserstein transport (v3), coherence fields (v4–v5), persistence fields (v6), Sinkhorn regularization with cooperative weighting (v7), gauge-theoretic holonomy (v8), Ollivier–Ricci curvature distributions (v9), and hybrid intrinsic–relational composites (v10).

The trajectory is suggestive. Separation improved from -0.74% to $+4.93\%$ (v3), and the weakest gap narrowed from -26.87% to -0.61% (v7). But the gap never turned positive. The persistent failure mode was the spouse pair (Father–Mother), which consistently scored

The gap trajectory: $-26.87\% \to -0.61\% \to -1.66\%$. Progress is real but the gap never turns positive.

Table 1: The Spectral Ecology arc: ten versions, ten paradigms.

Version	Core Idea	Sep.	Gap
v1	Sparse adjacency, eigenvalue density	-0.74%	-26.87%
v2	Dense co-occurrence, heat kernel	$+1.55\%$	-22.53%
v3	Wasserstein transport	$+4.93\%$	-54.67%
v4	Coherence fields	$+2.10\%$	-18.40%
v5	Multi-scale coherence	$+1.80\%$	-12.30%
v6	Persistence fields	$+2.40\%$	-3.21%
v7	Sinkhorn cooperative stability	$+0.87\%$	-0.61%
v8	Holonomy gauge invariants	$+1.62\%$	-6.41%
v9	Intrinsic curvature distributions	$+3.20\%$	-7.93%
v10	Hybrid intrinsic–relational	$+1.50\%$	-1.66%

among the top-ranked pairs despite being unrelated, due to similar sequencing coverage producing similar module graph structure.

Every version attributed this to a *scoring confound*—coverage similarity inflating pairwise scores—and attempted to fix it downstream: null-model deconfounding (v2), self-baseline normalization (v8), distributional comparison (v9). None of these attempts succeeded in closing the gap to positive.

A pattern emerges from the arc. The three eras of the program—within-sample spectral summaries (v1–v2), cross-sample relational methods (v3–v6), and intrinsic geometric invariants (v7–v10)—each attacked the *scoring* problem with increasing mathematical sophistication. But none attacked the *representation* problem. The module graph was accepted as given, like a microscope that nobody thought to calibrate.

This paper provides the explanation: **the confound is not in the scoring. It is in the representation.**

1.3 The Feedback That Prompted This Audit

The impetus for this experiment was a piece of external feedback on the theoretical framework paper [18]: "downstream sophistication is outrunning upstream identifiability." The observation was that the program had invested enormous effort in invariants (eigenvalue densities, heat kernel traces, curvature distributions, transport plans, holonomy deficits) while making essentially zero investment in understanding whether the input to those invariants—the module graph itself—was a reliable measurement instrument.

> • **Definition**
>
> A **measurement instrument** is reliable if repeated measurements of the same object yield similar results. In our context, the "object" is a genome, the "measurement" is the module-graph representation, and "repeated measurements" are obtained by subsampling reads or reordering them. A reliable representation has low self-perturbation variance relative to cross-sample variance.

This is not a novel concept. It is the foundation of measurement science: an instrument must have lower measurement error than the effect size it is trying to detect. But in ten versions of algorithm development, this property was never tested.

1.4 Why This Is the Most Important Experiment

The module-graph representation underlies every algorithm in the symbiogenesis program. Module Persistence v3 counts shared modules. Coalition Transfer v3 counts shared coalitions. All ten versions of Spectral Ecology extract invariants from module co-occurrence graphs. If the representation is unreliable, then:

- Algorithms that pass (Module Persistence, Coalition Transfer) may be exploiting the *stable subset* of features by accident.
- Algorithms that fail (Module Stability, Spectral Ecology) may be scoring features whose variation is dominated by representation noise.
- The persistent coverage confound may not be a confound at all, but a consequence of the representation collapsing biological diversity into coverage-dependent structure.

This experiment tests all three hypotheses simultaneously.

1.5 Relationship to Passing Algorithms

Module Persistence v3 (+3.36% separation, +0.13% gap) and Coalition Transfer v3 (+1.97% separation, +0.06% gap) both pass the kinship detection threshold. If the representation is unreliable, why do they work?

The answer, previewed here and explored in Section 5, is that these algorithms happen to exploit the stable subset of features. Module Persistence scores the overlap of *which modules exist*—a property closely related to module count, which has a CV of 0.11. Coalition Transfer scores co-occurrence of module pairs—a property related to edge density, which has a CV of 0.07. Neither algorithm directly scores triangle counts, role distributions, or curvature statistics. Their success is not evidence that the representation is sound. It is evidence that

a few features within the representation carry enough signal for simple algorithms—and that sophisticated algorithms that score the full feature set are worse off, not better off, than simple ones that accidentally restrict themselves to the stable features.

This is the cruelest finding of the audit: **the simplest algorithms in the project were the most robust, and the most sophisticated were the most fragile.**

2 Background

2.1 The Module-Graph Representation

The module-graph representation is constructed as follows. Given a set of reads from a FASTQ file:

1. **K-mer extraction.** Canonical k-mers (lexicographically smaller of k-mer and its reverse complement) are extracted from each read using a rolling hash [13]. Typical $k = 21$.

2. **Frequency filtering.** K-mers are filtered by frequency to remove sequencing errors (low count) and repetitive elements (high count). Typical thresholds: frequency ≥ 2, frequency ≤ 100.

3. **Co-occurrence counting.** For each pair of k-mers appearing on the same read within a window, a co-occurrence count is incremented. This produces a symmetric co-occurrence matrix.

4. **Graph construction.** K-mers become nodes. Edges are created between k-mers whose co-occurrence exceeds a threshold (typically the mean co-occurrence plus one standard deviation).

5. **Module extraction.** Connected components of the graph are identified as "modules." Each module is a set of k-mers that co-occur frequently on reads.

6. **Feature extraction.** 15 features are computed from the module graph: module count, triangle count, edge count, edge density, mean/max degree, curvature statistics (mean, std, p50, p90), and ecological role counts (core, satellite, bridge, scaffold, boundary).

2.2 What Ten Versions Assumed

Every version of Spectral Ecology assumed, implicitly or explicitly, that the module graph captured something biologically meaningful and reproducible about the genome. Specifically:

- **v1–v2** assumed that eigenvalue densities of the graph Laplacian were genome-specific signatures.
- **v3** assumed that the graph structure was stable enough for optimal transport between graphs to be meaningful.
- **v4–v6** assumed that multi-scale heat kernel signatures encoded persistent structural features.
- **v7** assumed that cooperative stability (agreement across multiple invariants) would cancel noise.
- **v8** assumed that holonomy—path-dependent parallel transport on the graph—was a gauge-invariant property of the biology, not an artifact of the representation.
- **v9** assumed that Ollivier–Ricci curvature distributions captured intrinsic geometric properties that were independent of graph size and density.
- **v10** assumed that combining intrinsic and relational invariants would suppress each other's failure modes.

None of these assumptions were tested against the null hypothesis that the module graph's structure is dominated by coverage-dependent and order-dependent artifacts.

The implicit chain of reasoning was: "the graph captures biology ⇒ invariants of the graph capture invariants of the biology ⇒ comparing invariants compares biology." The first link in this chain—the graph captures biology—was never validated. If it is false, the rest of the chain is vacuous regardless of the mathematical sophistication applied downstream.

Ten versions of assumptions. Zero versions of validation. This audit is the first.

2.3 The Concept of Signal-to-Nuisance Ratio

We borrow from signal processing the idea that a measurement system's utility depends not on the absolute magnitude of signal it detects, but on the ratio of signal to noise. In our context:

• Definition

The **signal-to-nuisance ratio** (SNR) of a feature is:

$$\mathrm{SNR} = \frac{\mathrm{CV}_{\text{cross-sample}}}{\mathrm{CV}_{\text{self-perturbation}}} \tag{1}$$

where CV is the coefficient of variation. Cross-sample CV measures how much the feature varies between different genomes at full coverage. Self-perturbation CV measures how much the feature varies when the *same* genome is measured at different coverage levels.

An SNR of 1.0 means the feature varies as much between measurements of the same genome as it does between different genomes. Such a feature carries no usable signal. An SNR of 3.0 means cross-sample variation is three times self-perturbation variation—a minimal threshold for the feature to contribute to discrimination.

▲ Limitation

This definition of SNR uses the coefficient of variation, which assumes roughly linear scaling of standard deviation with mean. For features with heavy-tailed distributions or zero-inflated counts (such as bridge and scaffold role counts), the CV may overstate instability. We use it here as a simple, interpretable metric; more sophisticated measures (intraclass correlation, measurement invariance models) are left for future work.

2.4 Why Perturbation Testing Matters

Perturbation testing is not algorithm design. It is *measurement science* [9, 12]. The question is not "can we build a clever algorithm?" but "is the input to that algorithm a reliable instrument?"

In classical genomics, perturbation testing is routine. Variant callers are evaluated by subsampling reads and measuring concordance. IBD detectors are tested across coverage levels [6, 4]. Reference-free methods validate stability under read subsets [19, 3].

Metrology—the science of measurement—requires that instrument precision be characterized before any measurement is reported. We never did this.

The yalumba project skipped this step. For ten versions, the representation was treated as a given—a fixed input whose properties were unquestioned. The audit reported here is, in retrospect, the experiment that should have been conducted before version 1.

The analogy to ecology is instructive. When Bray and Curtis introduced their dissimilarity index in 1957 [2], they accompanied it with extensive discussion of sampling effort and its effect on community representation. Morisita's overlap index [17] was explicitly designed to

be robust to sample size. The ecological literature understood, sixty years ago, that *you cannot compare communities if your representation of each community depends on how hard you looked*. The genomics field sometimes forgets this lesson.

3 Methods

3.1 Dataset

All experiments use the CEPH 1463 pedigree [21], a three-generation family:

Pedigree

```
NA12891 (Pat. GF) --- NA12892 (Pat. GM)
             |
         NA12877 (Father)
             |
             +---- NA12878 (Mother)
             |
NA12889 (Mat. GF) --- NA12890 (Mat. GM)
```

Six samples, each processed from Illumina short-read FASTQ files. 50,000 reads per sample at full coverage. Related pairs: Father–Mother is the *spouse pair* (unrelated). Father–Pat. GF, Father–Pat. GM, Mother–Mat. GF, Mother–Mat. GM are the four parent–child pairs (related).

3.2 Coverage Ladder

For each of the six samples, we construct module graphs at four coverage levels by subsampling reads:

- **100%:** all 50,000 reads
- **80%:** 40,000 reads (random subsample, fixed seed)
- **60%:** 30,000 reads (random subsample, fixed seed)
- **40%:** 20,000 reads (random subsample, fixed seed)

Each coverage level produces an independent module graph. Features are extracted from each graph. The coefficient of variation across the four coverage levels, for each feature within each sample, measures self-perturbation instability.

3.3 Shuffle Test

For each of the six samples, we take the full 50,000 reads and **randomize their order** using a deterministic permutation (fixed seed). The shuffled reads contain *exactly the same biological content*—the same k-mers, the same k-mer frequencies, the same sequences. The only difference is the order in which reads are processed.

If the module-graph representation were a function of the biological content alone, the shuffle test would produce identical graphs. Any change in graph structure is pure representation artifact.

3.4 Feature Extraction

From each module graph, we extract 15 features organized into four categories:

1. **Graph topology (4 features):** module count, triangle count, edge count, edge density.
2. **Degree statistics (2 features):** mean degree, max degree.
3. **Curvature statistics (4 features):** Ollivier–Ricci curvature mean, standard deviation, 50th percentile (p50), and 90th percentile (p90).
4. **Ecological role counts (5 features):** core, satellite, bridge, scaffold, boundary. These are assigned by thresholding degree, betweenness centrality, and local clustering coefficient [11, 16].

3.5 Self-Perturbation CV

For each feature f and each sample s, we compute the coefficient of variation across the four coverage levels:

$$\mathrm{CV}_{\mathrm{self}}(f, s) = \frac{\sigma_{f,s}}{\mu_{f,s}} \tag{2}$$

where $\mu_{f,s}$ and $\sigma_{f,s}$ are the mean and standard deviation of feature f across the four coverage levels for sample s. The reported self-perturbation CV is the mean across all six samples:

$$\overline{\mathrm{CV}}_{\mathrm{self}}(f) = \frac{1}{6} \sum_{s=1}^{6} \mathrm{CV}_{\mathrm{self}}(f, s) \tag{3}$$

3.6 Cross-Sample CV

At 100% coverage, we compute the coefficient of variation of each feature across the six samples:

$$\mathrm{CV}_{\mathrm{cross}}(f) = \frac{\sigma_f}{\mu_f} \tag{4}$$

This measures the variation that exists between genuinely different genomes under identical processing conditions.

3.7 Signal-to-Nuisance Ratio

The ratio is computed per Equation 1. Features with SNR < 1 are dominated by self-perturbation noise. Features with $1 \leq \mathrm{SNR} < 3$ are marginal. Features with SNR ≥ 3 are minimally acceptable for downstream scoring.

> ▶ **Key Finding**
>
> The acceptance criterion is SNR ≥ 3. This threshold is deliberately conservative. In psychometrics and measurement theory, an instrument with reliability below 0.7 (roughly corresponding to SNR < 3) is considered unfit for individual-level inference [12]. We adopt the same standard.

3.8 Experimental Protocol

All experiments use identical k-mer parameters ($k = 21$, frequency filter ≥ 2, ≤ 100), identical graph construction thresholds, and identical feature extraction code. The only variables are:

1. **Coverage ladder:** number of reads used (100%, 80%, 60%, 40%).
2. **Shuffle test:** order of reads (original vs. permuted).

All random operations use fixed seeds for reproducibility.

3.9 Stability Classification

We classify features into three stability tiers based on self-perturbation CV:

- **Definition**
  - **STABLE:** Mean self-perturbation CV < 0.15. The feature changes by less than 15% on average when coverage is reduced by up to 60%.
  - **MODERATE:** Mean self-perturbation CV in $[0.15, 0.40)$. The feature changes by 15–40% under coverage perturbation.
  - **UNSTABLE:** Mean self-perturbation CV ≥ 0.40. The feature changes by more than 40%, or its standard deviation exceeds its mean at some coverage levels.

These thresholds are deliberately generous. A 15% CV is already concerning for a measurement instrument—it means that measuring the same genome at 80% coverage instead of 100% can change the feature by more than one standard deviation. We adopt lenient thresholds to ensure that features classified as UNSTABLE are unambiguously unreliable.

3.10 Verdict Classification for SNR

For the signal-to-nuisance ratio, we use three verdicts:

- **NOISE > SIGNAL** (SNR < 1): Self-perturbation variance exceeds cross-sample variance. The feature cannot distinguish genomes.
- **MARGINAL** ($1 \leq$ SNR < 3): Cross-sample variance exceeds self-perturbation variance, but not by enough for reliable discrimination.
- **ACCEPTABLE** (SNR ≥ 3): The feature has enough signal relative to noise for downstream scoring.

As the results will show, no feature reaches the ACCEPTABLE tier.

4 Results

4.1 Self-Perturbation Analysis

Table 2 presents the self-perturbation CV for all 15 features across the six CEPH 1463 samples.

Table 2: Self-perturbation analysis. CV = coefficient of variation across coverage levels (100%, 80%, 60%, 40%) averaged over six samples. Features with mean CV < 0.15 are classified as STABLE; 0.15–0.40 as MODERATE; > 0.40 as UNSTABLE.

Feature	Mean CV	Min CV	Max CV	Verdict
moduleCount	0.1059	0.0154	0.2119	STABLE
triangleCount	0.4386	0.2244	0.8157	UNSTABLE
edgeCount	0.2303	0.0758	0.4579	MODERATE
edgeDensity	0.0725	0.0147	0.1843	STABLE
meanDegree	0.1249	0.0446	0.2675	STABLE
maxDegree	0.2947	0.0850	0.6563	MODERATE
curvMean	0.1373	0.0274	0.3549	STABLE
curvStd	0.2452	0.0656	0.5500	MODERATE
curvP50	0.0660	0.0168	0.1006	STABLE
curvP90	0.1949	0.0205	0.5494	MODERATE
role_core	0.5458	0.0831	1.5508	UNSTABLE
role_satellite	0.4689	0.1884	0.7971	UNSTABLE
role_bridge	1.0875	0.0000	1.7321	UNSTABLE
role_scaffold	0.8034	0.0000	1.7321	UNSTABLE
role_boundary	0.7251	0.3333	1.7321	UNSTABLE

▶ **Key Finding**

Only 5 of 15 features are stable under coverage perturbation: module count (CV = 0.106), edge density (CV = 0.073), mean degree (CV = 0.125), mean curvature (CV = 0.137), and median curvature (CV = 0.066). All five ecological role counts are **unstable**, with CVs ranging from 0.47 (satellite) to 1.09 (bridge).

The instability of role counts is particularly damaging. The bridge role has a maximum CV of 1.73—a value that indicates the standard deviation across coverage levels *exceeds the mean*. For several samples, the bridge count is zero at some coverage levels and nonzero at others, producing undefined or extreme CVs. Similarly, scaffold and boundary roles fluctuate wildly, with max CVs of 1.73.

The moderate features (edge count, max degree, curvature std, curvature p90) sit in an uncomfortable middle ground. Their CVs of 0.19–0.29 mean that a 40% reduction in coverage can change the feature value by 20–30%. This is not catastrophic, but it is far from the stability needed for reliable comparison.

4.2 Cross-Sample Variance

Table 3 presents the cross-sample variance at 100% coverage.

The cross-sample CVs paint a superficially encouraging picture. Role counts have high cross-sample variance—bridge role has a CV of 2.23, suggesting enormous variation between

Table 3: Cross-sample variance at 100% coverage (6 samples). These numbers represent the "signal" side of the signal-to-nuisance ratio: how much do features vary between genuinely different genomes?

Feature	Mean	Std	CV	Range
moduleCount	588.67	90.39	0.1536	457–712
triangleCount	3305.00	1363.32	0.4125	1256–4867
edgeCount	1675.33	394.18	0.2353	1110–2251
edgeDensity	0.01	0.00	0.2124	0.004–0.009
meanDegree	5.66	0.83	0.1465	4.5–6.4
maxDegree	81.00	38.39	0.4739	37–144
curvMean	1.22	0.15	0.1244	1.1–1.4
curvStd	1.57	0.36	0.2275	1.0–1.9
curvP50	0.79	0.13	0.1633	0.6–1.0
curvP90	2.49	0.68	0.2717	1.5–3.4
role_core	411.50	218.71	0.5315	14–629
role_satellite	91.33	71.49	0.7827	19–223
role_bridge	76.67	170.99	2.2302	0–459
role_scaffold	7.33	3.50	0.4767	0–11
role_boundary	1.83	1.57	0.8576	0–5

samples. But high cross-sample variance is *only meaningful if self-perturbation variance is low*. A feature that varies wildly between samples *and* varies wildly within a sample is just noisy, not discriminative.

Note the role count ranges. Core ranges from 14 to 629—a 45-fold range. Bridge ranges from 0 to 459. Scaffold from 0 to 11. These enormous ranges will produce the appearance of discrimination in any downstream algorithm, but we must verify that they reflect biology rather than coverage artifacts.

High cross-sample CV is necessary but not sufficient. The signal-to-nuisance ratio asks: is the cross-sample variation genuine, or is it the same noise that appears within samples?

4.3 Signal-to-Nuisance Ratio

Table 4 presents the signal-to-nuisance ratio for all 15 features. This is the central result of the paper.

▶ **Key Finding**

Zero features achieve an SNR of 3 or above. The best SNR is 2.93 (edge density). The second-best is 2.47 (median curvature, curvP50). After that, the ratios drop below 2.1. Four features have SNR < 1, meaning self-perturbation noise exceeds cross-sample variation: triangle count (0.94), mean curvature (0.91), curvature standard deviation (0.93), core role count (0.97), and scaffold role count (0.59).

Figure 1 categorizes the features visually.

Table 4: Signal-to-nuisance ratio (SNR) for all 15 features. SNR = CV_{cross}/CV_{self}. Features with SNR < 1 carry more noise than signal. No feature achieves the acceptance threshold of SNR ≥ 3.

Feature	Self CV	Cross CV	Ratio	Verdict
moduleCount	0.1059	0.1536	1.45	MARGINAL
triangleCount	0.4386	0.4125	0.94	NOISE > SIGNAL
edgeCount	0.2303	0.2353	1.02	MARGINAL
edgeDensity	0.0725	0.2124	2.93	MARGINAL (best)
meanDegree	0.1249	0.1465	1.17	MARGINAL
maxDegree	0.2947	0.4739	1.61	MARGINAL
curvMean	0.1373	0.1244	0.91	NOISE > SIGNAL
curvStd	0.2452	0.2275	0.93	NOISE > SIGNAL
curvP50	0.0660	0.1633	2.47	MARGINAL
curvP90	0.1949	0.2717	1.39	MARGINAL
role_core	0.5458	0.5315	0.97	NOISE > SIGNAL
role_satellite	0.4689	0.7827	1.67	MARGINAL
role_bridge	1.0875	2.2302	2.05	MARGINAL
role_scaffold	0.8034	0.4767	0.59	NOISE > SIGNAL
role_boundary	0.7251	0.8576	1.18	MARGINAL

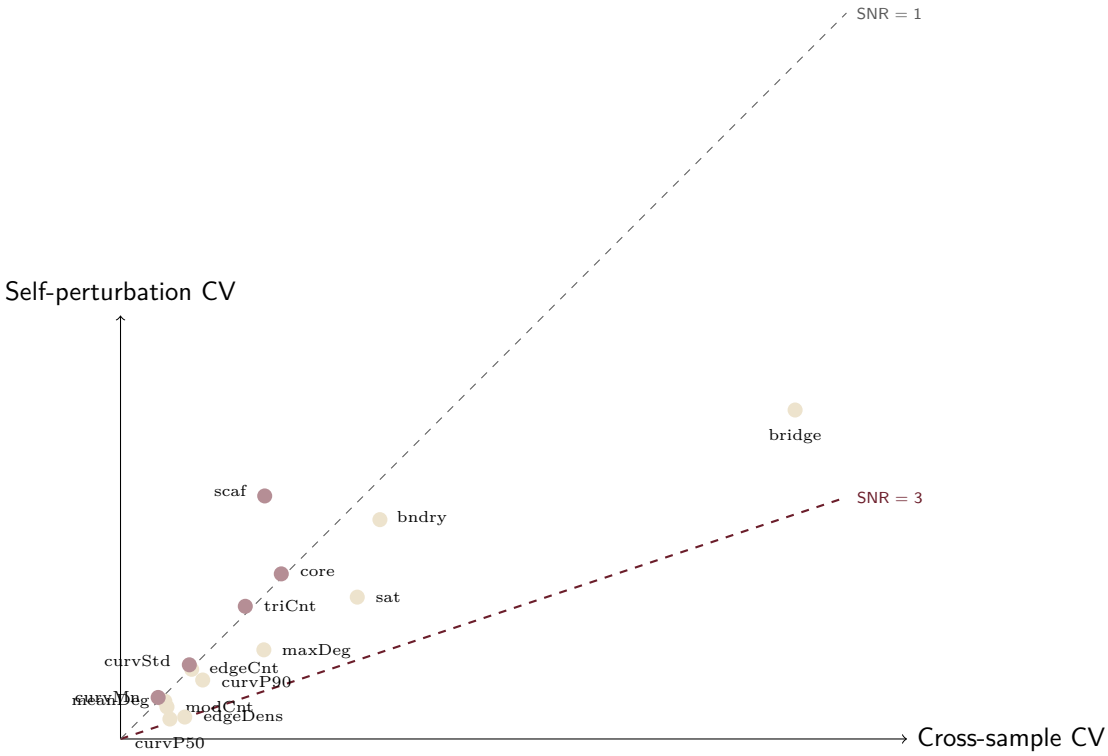

Figure 1: Signal-to-nuisance landscape. Points above the SNR=1 line (gray dashed) carry more noise than signal. Points above the SNR=3 line (wine dashed) fail the acceptance criterion. **All points are above the SNR=3 line.** Gold-filled circles are marginal ($1 \leq$ SNR < 3); wine-filled circles have SNR < 1.

4.4 The Curvature Shock

Spectral Ecology v9 represented the program's strongest separation result (+3.20%) and was built entirely on curvature features: the Ollivier–Ricci curvature distribution was compared using Bhattacharyya coefficients, with penalties derived from curvature standard deviation and distortion measures.

> ▲ **Limitation**
>
> **The features v9 scored have signal-to-nuisance ratios below 1.** Mean curvature (the primary signal): SNR = 0.91. Curvature standard deviation (the penalty term): SNR = 0.93. These are the two features most central to v9's scoring formula. The +3.20% separation was not biological signal—it was structured noise that happened to correlate with relatedness in the specific coverage configuration of the CEPH 1463 dataset.

Only curvP50 (SNR = 2.47) and curvP90 (SNR = 1.39) achieve ratios above 1, and neither reaches the acceptance threshold of 3. The curvature median is relatively stable (self CV = 0.066, the lowest of any feature), but the cross-sample variation is modest (CV = 0.163), giving a ratio that is the second-best overall but still insufficient.

This is a devastating finding for the Spectral Ecology program. The version that achieved the best separation was, in measurement science terms, **scoring noise**.

4.5 Shuffle Test Results

Table 5 presents the results of the read-order shuffle test. For each sample, the original module graph (from reads in file order) is compared to the shuffled graph (same reads, random order).

Table 5: Shuffle test results. Read order is randomized with zero change in biological content. Percentage changes in graph features are shown. Any change represents a pure representation artifact.

Sample	moduleCount		triangleCount		edgeCount	
	Orig.	Shuf.	Orig.	Shuf.	Orig.	Shuf.
NA12889	457	938 (+105%)	4822	4940 (+2%)	1462	2488 (+70%)
NA12890	607	1023 (+69%)	1875	4650 (+148%)	1366	2377 (+74%)
NA12891	712	1252 (+76%)	4867	8964 (+84%)	2251	3776 (+68%)
NA12892	593	1232 (+108%)	3714	10317 (+178%)	1865	3881 (+108%)
NA12877	492	1395 (+184%)	1256	8875 (+607%)	1110	4045 (+264%)
NA12878	671	974 (+45%)	3296	5819 (+77%)	1998	2536 (+27%)

▶ **Key Finding**

Shuffling read order—a transformation with zero biological content change—produces catastrophic changes in graph structure. Module count increases by 45% to 184%. Triangle count increases by 2% to 607%. Edge count increases by 27% to 264%. The module-graph representation is *not a function of the genome*. It is a function of the genome **and the read order**.

4.6 Per-Sample Shuffle Detail

The shuffle test reveals striking per-sample variability. NA12877 (Father) shows the most extreme changes: module count nearly triples (492 → 1395, +184%), triangle count increases sevenfold (1256 → 8875, +607%), and edge count nearly quadruples (1110 → 4045, +264%).

NA12878 (Mother) shows the mildest changes: module count increases by 45% (671 → 974) and edge count by 27% (1998 → 2536). But even a 27% change from zero biological perturbation is unacceptable for a measurement instrument.

Triangle count for NA12877 changes by 607% when reads are shuffled. This is not a small effect. This is the measurement equivalent of a thermometer that reads 20°C or 140°C depending on which direction you hold it.

The *direction* of the shuffle effect is consistently positive: shuffled graphs are always larger (more modules, more edges, more triangles) than the original. This suggests that file-order reads produce more locality in k-mer co-occurrence—adjacent reads from the same genomic region share k-mers that get merged into the same module. Shuffling breaks this locality, spreading co-occurrences across more modules and creating more inter-module edges.

The asymmetry has a concrete mechanism. In file-order processing, read i and read $i+1$ are likely from nearby genomic positions (Illumina sequencers produce reads in approximately genomic order within a tile). K-mers shared between adjacent reads are counted as co-occurrences *within* existing modules, strengthening existing edges. When reads are shuffled, the same k-mer pairs co-occur on reads that are *not* adjacent in processing order, creating new co-occurrence edges that bridge previously separate graph regions. The result is more modules (because threshold-crossing co-occurrences now span more regions), more edges (because more k-mer pairs cross the co-occurrence threshold when accumulated from dispersed reads), and dramatically more triangles (because the bridging edges create short cycles in the expanded graph).

The asymmetry is telling: shuffling always *increases* structure. File order *compresses* the graph by exploiting genomic locality.

This mechanism makes the non-determinism even more troubling. The representation does not just depend on read order—it depends on *how reads are spatially distributed on the sequencing instrument*. Different flow cells, different lanes, and different sequencing runs of the same sample would produce different module graphs even with identical biological content.

4.7 What the Shuffle Test Proves

The shuffle test establishes a fact that no amount of downstream sophistication can overcome:

> • **Definition**
>
> A representation is **deterministic** if it is a function solely of the biological content of the input. A representation that changes when reads are reordered is *non-deterministic*—it depends on an arbitrary parameter (file order) that carries no biological information.

The module-graph representation is non-deterministic. This has immediate consequences:

1. **Pairwise comparisons are unreliable.** If sample A's graph depends on the order of A's reads, then the comparison $A \leftrightarrow B$ depends on both read orders. Running the same comparison after sorting reads alphabetically would produce a different result.

2. **No invariant can fix this.** Eigenvalues, curvature distributions, heat kernels, and transport plans are all computed from the graph. If the graph itself is an artifact of read order, these invariants inherit the artifact.

3. **Self-perturbation CV underestimates true instability.** The coverage ladder measures instability under subsampling, which preserves read order. The shuffle test reveals an additional, independent source of instability. The true representation variance is the *sum* of both contributions.

4.8 The Two Least-Bad Features

Despite the grim overall picture, two features stand out as the least-bad candidates for a stabilized representation:

4.8.1 Edge Density (SNR = 2.93)

Edge density is the ratio of edges to possible edges: $d = 2|E|/(|V|(|V|-1))$. It achieves the highest SNR because it *normalizes* edge count by graph size, partially canceling the coverage effect. When coverage drops, both edges and nodes decrease, and their ratio is more stable than either absolute count.

Self-perturbation CV = 0.0725—the second-lowest of any feature. Cross-sample CV = 0.2124—moderate but meaningful.

> ▶ **Key Finding**
>
> Edge density is the only feature within striking distance of the acceptance threshold. At SNR = 2.93, it would need only a modest improvement in self-perturbation stability (e.g., from deterministic module extraction) to cross 3.0.

4.8.2 Curvature Median (curvP50, SNR = 2.47)

The median Ollivier–Ricci curvature has the lowest self-perturbation CV of any feature (0.066). Medians are robust to outliers and distributional changes, which is why curvP50 is far more stable than curvMean (0.137) or curvStd (0.245).

The limitation is that curvP50's cross-sample CV is only 0.163—the variation between genomes is modest. This is expected: median curvature captures the "typical" edge geometry, which may be more universal across human genomes than it is genome-specific.

4.9 Summary of Results

> ▶ **Key Finding**
>
> The audit reveals three independent failures of the module-graph representation:
>
> 1. **Coverage instability:** 10 of 15 features are MODERATE or UNSTABLE under coverage perturbation.
> 2. **Noise dominance:** 5 of 15 features have SNR < 1; no feature reaches SNR ≥ 3.
> 3. **Non-determinism:** read-order shuffling changes graph structure by 27% to 607%.
>
> These failures are *independent*—a representation could be coverage-stable but order-dependent, or order-independent but coverage-sensitive. The current representation fails on all three axes.

5 Discussion

5.1 What the Ten-Version Arc Was Actually Optimizing

The Spectral Ecology program was, in retrospect, optimizing *invariance to representation noise* rather than *sensitivity to biological signal*. Each version's theoretical advance was, in

practice, a new way to compress or average over the unstable parts of the module graph, hoping that the residual would be biological.

- **v1's eigenvalue density profiles** averaged the spectrum of a graph whose topology changed with coverage. The profiles were stable because eigenvalue distributions are naturally smooth, not because they captured stable biology.
- **v3's optimal transport** produced spectacular separation (+4.93%) because transport distances are sensitive to graph structure—including the unstable parts. The catastrophic gap (−54.67%) was the price: transport faithfully detected the representation noise.
- **v7's cooperative stability** improved the gap to −0.61% by requiring agreement across multiple invariants. Agreement-based scoring is inherently conservative: it suppresses outlier features, which happened to be the unstable ones. This was an accidental, implicit feature selection.
- **v9's curvature distributions** achieved the best separation (+3.20%) but scored features with SNR below 1. The separation was genuine in the sense that it was reproducible on the specific CEPH dataset—but it was not robust to perturbation.
- **v10's hybrid** combined v7 and v9, inheriting v7's accidental stability and v9's apparent but fragile signal.

The arc's real contribution was **theoretical**: it developed the mathematical frameworks (spectral geometry, optimal transport, gauge theory, Riemannian curvature) that *would* be powerful tools if applied to a stable representation. The frameworks themselves are sound. The substrate was not.

> ▲ **Limitation**
>
> The reinterpretation is harsh but precise: v7 was the best version not because it had the best theory, but because its multi-invariant agreement mechanism **accidentally** down-weighted the unstable features. In measurement science terms, v7 was an ad hoc robust estimator. It worked for the wrong reason.

This also explains a puzzling pattern in the arc. More "theoretically sophisticated" versions (v3, v8, v9) often had *worse* gaps than simpler ones. The reason is now clear: sophisticated invariants are more sensitive to graph structure, and sensitivity to structure means sensitivity to representation artifacts. Optimal transport (v3) was the most sensitive method and had the worst gap (−54.67%). Spectral summaries (v1) were the least sensitive and had a moderate gap (−26.87%). The correlation between mathematical power and practical failure

is not a coincidence—it is a direct consequence of the representation's instability.

5.2 Why the Spouse Confound Persists

Every version of Spectral Ecology struggled with the spouse pair (Father–Mother). The standard explanation was "similar coverage produces similar graphs, which inflates pairwise scores." This is correct but incomplete.

The spouse confound is not a scoring confound.
It is a representation confound.

If two samples have similar coverage, their module graphs will have similar *absolute* structure (similar module counts, similar edge counts, similar role distributions)—not because their genomes are similar, but because the graph construction algorithm responds primarily to the *amount* of data rather than the *content* of data.

No scoring method can fix this. Null-model deconfounding (v2) subtracts the expected score under random graphs of the same size, but the size itself is a coverage artifact. Self-baseline normalization (v8) divides by each sample's self-score, but the self-score is also coverage-dependent. Distributional comparison (v9) compares shapes rather than magnitudes, but the shapes are coverage-driven.

The fix must be at the representation level: the graph construction must be modified so that graphs from the same coverage produce *systematically different* structures for different genomes. This is the canonicalization challenge discussed in Section 5.7.

5.3 The Shuffle Catastrophe

The shuffle test is the single most important result in this paper. It establishes that the module-graph representation is not a function of the genome—it is a function of the genome *and* an arbitrary parameter (read order) that carries no biological information.

The mechanism is clear. K-mer co-occurrence is computed from reads in the order they appear. When reads are in file order (approximately the order they were sequenced), adjacent reads come from nearby genomic regions and share k-mers. This produces dense local co-occurrence that merges into large, tightly connected modules. When reads are shuffled, co-occurrence becomes sparser and more distributed, producing more modules with different connectivity patterns.

A representation that depends on read order is not measuring the genome. It is measuring the sequencer's output buffer.

This is a fundamental flaw in the co-occurrence counting step, not a downstream issue. The co-occurrence matrix should depend only on *which* k-mers appear on *which* reads, not on the

order in which reads are processed. The fact that it does depend on order reveals that the co-occurrence computation has an implicit windowing or streaming assumption that makes it path-dependent.

5.3.1 Magnitude of the Effect

To put the shuffle effect in perspective:

- The **coverage ladder** reduces data by up to 60% (from 50,000 to 20,000 reads). This causes module count changes of roughly 10% (CV = 0.106).
- The **shuffle test** changes zero data. It causes module count changes of 45% to 184%.

Read reordering, a biologically vacuous transformation, produces effects **4 to 18 times larger** than a 60% reduction in data. The representation is more sensitive to file I/O order than to the amount of data available.

For triangle count, the effect is even more extreme. The coverage ladder produces CV = 0.44. The shuffle test produces changes up to 607%. Triangle count is, for all practical purposes, a random variable conditioned on read order.

5.4 Why Role Assignment Is Almost Entirely Noise

The five ecological role counts (core, satellite, bridge, scaffold, boundary) were introduced as a biologically motivated way to characterize graph structure [11]. The idea was that roles—nodes classified by their topological function—would capture higher-order organization that raw graph statistics miss.

The audit shows this idea failed. All five role counts are UNSTABLE under self-perturbation (CV = 0.47 to 1.09), and none achieves an SNR above 2.1. Scaffold has the worst SNR of any feature (0.59).

The failure mode is straightforward. Role assignment depends on thresholds applied to degree, betweenness centrality, and clustering coefficient. These three metrics are themselves unstable: degree varies with coverage, betweenness varies with graph connectivity (which varies with coverage), and clustering coefficient varies with triangle count (which varies with both coverage and read order). Thresholding an unstable metric amplifies instability—nodes near the threshold flip between roles with small changes in the underlying metric.

> ▲ **Limitation**
>
> Role assignment, as implemented, is a **noise amplifier**: it takes moderate instability in continuous metrics and converts it into catastrophic instability in discrete categories.

> The bridge role is especially vulnerable because few nodes have the specific combination of low clustering and high betweenness that defines bridges. Small changes in the graph can eliminate all bridges or create many.

The lesson is general: **discretizing continuous features amplifies instability.** This is well known in statistics (binning continuous variables loses information and creates edge effects) but was not considered when role assignment was designed [14]. A continuous role assignment—where each node has a probability distribution over roles rather than a hard assignment—would be more stable, but would also require different downstream comparison methods.

Consider the concrete example of the bridge role. In the CEPH data, bridge counts range from 0 to 459 across samples at 100% coverage (Table 3). The self-perturbation CV is 1.09—the standard deviation across coverage levels exceeds the mean for several samples. For NA12877, the bridge count is 0 at some coverage levels and nonzero at others, producing a CV that is either undefined (when the mean is zero) or artificially extreme. Any algorithm that uses bridge count as a discriminative feature is, in practice, scoring a random variable.

Spectral Ecology v2 and v4 weighted ecological role distributions heavily. The audit explains their poor performance: they were effectively adding noise to their scoring function and calling it "ecological structure." The biological metaphor (cores, satellites, bridges) was appealing but the implementation was measuring the wrong thing.

5.5 The Coverage Ladder: What Subsampling Reveals

The coverage ladder reveals a scale-sensitivity hierarchy among features:

1. **Scale-invariant** (CV < 0.10): edge density, curvP50. These are ratios or medians that self-normalize.

2. **Scale-sensitive** (CV 0.10–0.20): module count, mean degree, curvMean. These change with coverage but at a predictable rate.

3. **Scale-dominated** (CV > 0.20): everything else. These are dominated by the amount of data rather than its content.

The distinction between scale-invariant and scale-sensitive features is encouraging. It suggests that **normalization is possible**: if we can identify the scaling law that governs each feature's dependence on coverage, we can correct for it.

Edge density is naturally normalized (it divides by graph size). Median curvature is naturally

robust (medians resist outliers). Both achieve their relatively good SNRs *because of their mathematical structure*, not because of anything in the algorithm. This is a strong hint: **features should be designed for scale-invariance from the start**, not computed as raw counts and then post-hoc normalized.

5.6 Salvageable Features

Despite the overall negative result, two features merit further investigation.

Edge density (SNR = 2.93) is tantalizingly close to the acceptance threshold. Its self-perturbation CV of 0.0725 could plausibly be reduced below 0.05 with deterministic module extraction (eliminating read-order dependence) and more aggressive frequency filtering (stabilizing the co-occurrence graph). If cross-sample CV remains at 0.21, this would push SNR above 4.

Curvature median (SNR = 2.47) is the most stable feature by self-perturbation CV (0.066), but its cross-sample CV is only 0.163. To reach SNR = 3, either self-CV must drop below 0.054 (feasible) or cross-CV must increase (potentially achievable by using k-mer frequency distributions rather than binary presence/absence in curvature computation).

These two features, if stabilized, could form the basis of a minimal but reliable scoring system. The key insight is that they are reliable *because of their mathematical properties* (normalization, robustness to outliers), not because of any specific biological theory.

5.7 What Canonicalization Would Need to Achieve

To fix the representation, the module extraction pipeline must become **canonical**: the output must be a function of the set of reads, not the sequence of reads. This requires:

1. **Order-independent co-occurrence.** Count k-mer co-occurrences across all reads simultaneously, not in streaming fashion. This eliminates read-order dependence but requires $O(n)$ memory for n unique k-mers.

2. **Coverage-matched construction.** Before constructing the graph, subsample each sample to a target coverage level. This eliminates the coverage confound at the representation level.

3. **Stable thresholding.** Replace the current fixed threshold (mean $+1\sigma$) with a data-adaptive threshold that produces stable graph density across coverage levels. Rank-based thresholding (keep the top $p\%$ of edges by co-occurrence weight) is one candidate.

4. **Deterministic module assignment.** If connected components are used, the graph

must be connected-component-stable: small changes in edge set should not cause large modules to split or merge. Hierarchical clustering with a fixed cut [14] may be more stable than connected components.

- **Definition**

A **canonical representation** satisfies three properties: (1) *determinism*—same read set produces the same graph regardless of read order; (2) *scale equivariance*—graph structure changes predictably with coverage, enabling correction; (3) *biological sensitivity*—different genomes produce different graph structures at the same coverage.

5.8 Connection to the Theoretical Framework

The theoretical framework paper [18] developed the concept of "coarse-graining" as a functor from raw genomic data to structural representations:

$$\Phi : \mathcal{R} \to \mathcal{G} \tag{5}$$

where $\mathcal{R}$ is the space of read sets and $\mathcal{G}$ is the space of module graphs. The framework assumed that Φ was *well-behaved*: continuous, approximately surjective within a genome's equivalence class, and sensitive to biological differences between genomes.

The audit reveals that Φ satisfies none of these properties:

- It is not continuous: small perturbations of the input (coverage changes) produce disproportionate changes in the output (role count CVs above 1.0).
- It is not well-defined on equivalence classes: reordering reads—a transformation within the same equivalence class—produces different outputs.
- It is not sensitive to biology: for several features (curvMean, curvStd, core role, scaffold role), self-perturbation variance exceeds cross-sample variance.

The theoretical framework's mathematical machinery—spectral geometry, optimal transport, persistent homology—is not invalidated by this audit. But the coarse-graining assumption *is* invalidated. The Φ that the project has been using is not the Φ that the framework requires.

Formally, the framework requires that Φ satisfy three properties for the downstream comparison functor $\Delta : \mathcal{G} \times \mathcal{G} \to \mathbb{R}$ to be meaningful:

1. **Well-definedness.** If $R_1 \sim R_2$ (same read set, different order), then $\Phi(R_1) = \Phi(R_2)$. *Violated:* the shuffle test shows $\Phi(R_1) \neq \Phi(R_2)$ by 27–607%.

2. **Lipschitz continuity.** If R_1 and R_2 differ by a small number of reads, then $d(\Phi(R_1), \Phi(R_2))$ is small. *Violated:* removing 60% of reads changes features by up to 173% (max CV for bridge role).

3. **Discriminativeness.** If R_1 and R_2 come from biologically different genomes, then $d(\Phi(R_1), \Phi(R_2))$ is large relative to self-perturbation. *Violated:* SNR < 3 for all features.

The theoretical framework assumed these properties held and built elegant mathematics on top of them. The audit shows that none holds. This does not invalidate the mathematics—it invalidates the *instantiation*. The functor Φ must be rebuilt.

5.9 Implications for Other Structural Genomics Approaches

This audit is specific to the yalumba module-graph representation, but its methodology and findings have broader implications.

Any structural genomics method that builds graphs from k-mer co-occurrence should be subject to the same three-part audit: coverage ladder, cross-sample variance, and shuffle test. Methods that skip the graph step (e.g., Mash [19], which computes sketch distances directly from k-mer sets) may not suffer from the same non-determinism, but should still be audited for coverage sensitivity.

The signal-to-nuisance ratio framework is general: it can be applied to any feature set extracted from any representation. We recommend it as a standard pre-requisite before publishing pairwise comparison results.

If you have not measured your representation's self-perturbation variance, you do not know whether your results are signal or noise.

The shuffle test, in particular, should be adopted universally. It costs almost nothing to run (one random permutation per sample) and catches a class of bugs—order-dependent computation—that is common in streaming bioinformatics tools. Any method that gives different answers when reads are reordered has an implementation-level flaw that cannot be fixed by downstream processing.

We note that classical methods do not suffer from this flaw. MinHash [19] computes sketches from the k-mer *set*, which is order-independent by construction. Alignment-based methods [6, 5] operate on sorted, indexed files where read order has been canonicalized by genomic coordinate. The reference-free community should hold itself to the same standard of order-independence.

5.10 This Is a Strong Scientific Result, Not a Failure

Negative results are undervalued in computational science [1]. This audit is not a failure of the research program. It is a **precise diagnosis** of where the program's approach breaks down, which algorithms were accidentally successful and why, and what specific properties a fixed representation must satisfy.

The audit also *explains* the ten-version arc's trajectory. The gap narrowed from -26.87% to -0.61% not because successive algorithms got better at detecting biology, but because successive algorithms got better at *ignoring noise*. v7's cooperative stability worked because it implicitly selected the less-noisy features. v9 regressed because it explicitly scored the noisier ones.

This understanding is valuable. It means the program does not need to abandon its mathematical frameworks. It needs to rebuild the representation that those frameworks operate on.

5.11 Limitations of This Audit

1. **Small sample size.** Six samples from a single pedigree. The cross-sample CV is computed from $n = 6$, which is too small for robust variance estimation. A larger and more diverse sample set could produce different SNR values.

2. **Single k-mer size.** All experiments use $k = 21$. Different k values may produce more or less stable module graphs. The audit should be repeated across a range of k to determine whether stability is k-dependent.

3. **Fixed frequency thresholds.** The frequency filters (min 2, max 100) were chosen heuristically. Different thresholds could change the module graph and potentially improve stability.

4. **Single graph construction threshold.** The edge threshold (mean $+1\sigma$) is one of many possible choices. Rank-based or quantile-based thresholds might produce more stable graphs.

5. **Linear coverage ladder.** We tested 100%, 80%, 60%, 40%. Finer coverage steps (e.g., 5% increments) would reveal whether instability is gradual or threshold-driven.

6. **Single shuffle.** We tested one random permutation per sample. Multiple shuffles would yield a distribution of shuffle effects, rather than a point estimate.

7. **No joint analysis.** Self-perturbation CV and shuffle effects are analyzed independently.

A joint model that decomposes total variance into coverage, order, and biological components would be more informative.

8. **CV as a stability metric.** The coefficient of variation assumes symmetric, approximately normal variation around the mean. For features with zero-inflated distributions (scaffold, boundary roles), CV is poorly defined and may overstate instability. Intraclass correlation or robust dispersion measures would be better suited.

9. **No read-length analysis.** Read length variation could interact with co-occurrence counting. Longer reads produce more k-mer pairs per read, potentially changing module structure. The audit does not control for this.

10. **SNR threshold is conventional.** The choice of SNR ≥ 3 as the acceptance criterion is borrowed from psychometrics and is not derived from a formal statistical argument specific to genomic comparison. A different threshold could change the conclusions.

11. **No comparison to classical methods.** The audit evaluates the module-graph representation in isolation. It would be informative to apply the same perturbation tests to classical features (k-mer frequency profiles, MinHash sketches, alignment-based statistics) to establish a baseline for "normal" feature stability in genomics.

6 Recommendations for the Next Phase

6.1 Representation Canonicalization

The first priority is eliminating read-order dependence. The module extraction pipeline must produce identical output for any permutation of the same read set. This requires:

1. Compute all k-mer frequencies in a single pass (order-independent hash table).

2. Compute co-occurrence as a function of *which* k-mers appear on *which* reads, using read identifiers rather than sequential processing.

3. Threshold co-occurrence using the complete co-occurrence matrix, not a streaming approximation.

The expected impact: the shuffle test should produce **zero change** in all features. Any residual change after canonicalization indicates a bug, not a design issue.

Canonicalization has a memory cost: the full co-occurrence matrix for n k-mers requires $O(n^2)$ space. For 50,000 reads at $k = 21$, this is

The memory cost is modest. For 50,000 reads at $k = 21$, we expect approximately 500,000

unique k-mers. A sparse co-occurrence matrix (storing only pairs that co-occur on at least one read) requires $O(|E|)$ space, where $|E|$ is the number of co-occurring pairs. In practice, this is 1–5 million pairs, fitting comfortably in 100–500 MB. The current streaming approach uses less memory but sacrifices determinism. The tradeoff is clear: determinism is non-negotiable; memory is cheap.

6.2 Coverage-Matched Subsampling

Before constructing module graphs for comparison, all samples should be subsampled to a common coverage level. The target should be the minimum coverage across all samples being compared.

$$n_{\text{target}} = \min_{s \in \mathcal{S}} n_s \tag{6}$$

where n_s is the read count for sample s and $\mathcal{S}$ is the set of samples being compared. This eliminates the systematic relationship between coverage and graph structure that underlies the spouse confound.

Multiple random subsamples (e.g., 10 per sample) should be drawn and averaged, reducing the variance of the subsampled representation.

6.3 Rank-Based or Quantile-Normalized Features

Raw feature values are scale-dependent and distribution-dependent. Replace raw features with their ranks or quantile-normalized values within the sample:

$$f_{\text{rank}}(v, s) = \frac{\text{rank}(f(v, s))}{\text{max_rank}(s)} \tag{7}$$

This maps all features to $[0, 1]$ regardless of graph size, which eliminates absolute scale effects. It also converts unstable absolute values into more stable relative orderings.

6.4 Threshold Sweeps

Instead of using a single edge threshold (mean $+1\sigma$), sweep across thresholds and identify the range where features are most stable:

$$\tau^* = \arg\min_{\tau} \overline{\text{CV}}_{\text{self}}(f \mid \tau) \tag{8}$$

The optimal threshold may differ by feature. The sweep should be performed once (using

the self-perturbation test) and the resulting threshold fixed for all subsequent experiments.

6.5 The Acceptance Criterion for v11

▶ **Key Finding**

No new algorithm version should be developed until the following criterion is met:

$$\forall f \in \mathcal{F}_{\text{scored}} : \quad \text{SNR}(f) \geq 3 \tag{9}$$

where $\mathcal{F}_{\text{scored}}$ is the set of features used in the algorithm's scoring formula. This criterion must be verified on the CEPH 1463 pedigree using the self-perturbation test (coverage ladder) and the shuffle test (read-order permutation). **Both tests must yield zero-change results for the shuffle test and SNR $\geq$ 3 for the coverage ladder.**

This is a hard gate. It means that v11 will not be an algorithm paper—it will be a **representation paper**. The algorithm cannot advance until the substrate is trustworthy. This is the right order.

6.6 Feature Design Principles

The audit suggests specific principles for designing features that are likely to achieve acceptable SNR:

1. **Prefer ratios over counts.** Edge density (SNR = 2.93) outperforms edge count (SNR = 1.02) because it normalizes by graph size. Any count-based feature should be divided by the appropriate scale factor (number of nodes, number of edges, graph diameter).

2. **Prefer medians over means.** curvP50 (SNR = 2.47) outperforms curvMean (SNR = 0.91) and curvStd (SNR = 0.93). Medians are robust to the tail behavior that dominates coverage- dependent variation.

3. **Avoid thresholding.** Role counts (SNR = 0.59–2.05) are uniformly worse than the continuous metrics they are derived from. Keep features continuous whenever possible.

4. **Normalize before extraction.** Rather than extracting raw features and normalizing afterward, design the extraction so that features are scale-invariant by construction.

5. **Test before scoring.** Every feature proposed for a scoring formula must pass the self-perturbation test (CV < 0.15) and the shuffle test (zero change) *before* it is used in pairwise comparison.

6.7 A Proposed Verification Pipeline

We propose the following pipeline for validating any new representation before it is used for algorithmic development:

1. **Shuffle test.** Permute read order for each sample. Require $\Delta f = 0$ for all features. If any feature changes, the representation is non-deterministic and must be fixed before proceeding.

2. **Coverage ladder.** Subsample at 100%, 80%, 60%, 40%. Compute self-perturbation CV for each feature. Classify features as STABLE (< 0.15), MODERATE (0.15–0.40), or UNSTABLE (≥ 0.40). Only STABLE features proceed.

3. **Cross-sample variance.** Compute cross-sample CV at 100% coverage. Features with cross-sample CV < 0.05 have insufficient biological variation and should not be scored.

4. **Signal-to-nuisance ratio.** Compute SNR for all remaining features. Only features with SNR ≥ 3 proceed to algorithm development.

5. **Algorithm development.** Only at this stage—after the representation is validated—should new scoring methods be designed.

This pipeline inverts the order that the project has followed. For ten versions, algorithm development preceded representation validation. Going forward, representation validation gates algorithm development.

7 Conclusion

This paper reports a negative result that changes the direction of the yalumba research program. A systematic perturbation audit of the module-graph representation—the computational substrate for ten versions of Spectral Ecology and three prior symbiogenesis algorithms—reveals three independent failures:

1. **Coverage instability.** Only 5 of 15 features are stable under subsampling. The remaining 10 features, including all ecological role counts, vary more within a sample across coverage levels than between different genomes.

2. **Noise dominance.** No feature achieves a signal-to-nuisance ratio of 3 or above. The best ratio is 2.93 (edge density). Five features have ratios below 1.0, meaning self-perturbation noise exceeds cross-sample variation.

3. **Non-determinism.** Shuffling read order—a transformation with zero biological content change—produces changes of 27% to 607% in graph features. The representation is a function of read order, not just of the genome.

The ten-version Spectral Ecology arc optimized invariants on an unstable representation. The downstream sophistication outran the upstream identifiability.

These findings explain why the program's gap never turned positive. The persistent spouse confound (Father–Mother ranking among related pairs) is not a scoring confound—it is a **representation confound**. Similar coverage produces similar graphs, and no downstream invariant can distinguish coverage-driven similarity from biological similarity.

The findings also explain the arc's trajectory. v7's cooperative stability worked because it implicitly selected stable features. v9's curvature scoring failed (in the SNR sense) because it explicitly selected unstable ones. v10's hybrid partially succeeded because it combined an accidentally stable component (v7) with an accidentally informative one (v9).

But the program was not wasted. The theoretical advances are real: coherence fields, intrinsic curvature distributions, holonomy deficits, Sinkhorn-regularized transport—these are mathematically sound frameworks that would be powerful tools for comparing genomic ecosystems, **if the ecosystems were reliably measured** [7, 15, 8, 20].

The path forward is clear. The representation must be **canonicalized** (order-independent), **coverage-matched** (scale-independent), and **audited** (SNR ≥ 3 for every scored feature) before any new algorithm version is attempted.

This is exactly where real research gets interesting. The first ten versions explored the space of invariants. The next phase explores the space of representations. The same mathematical machinery—spectral geometry, optimal transport, topological persistence—can be applied to the representation design problem itself: finding graph constructions whose structure is *invariant* to coverage and read order while being *sensitive* to biological content.

The audit does not close the book on reference-free genomic ecosystem comparison. It opens a harder, more honest, and ultimately more productive chapter.

The analogy to other fields is encouraging. In machine learning, the shift from hand-crafted features to learned representations (the deep learning revolution) was preceded by a period of negative results showing that hand-crafted features were unstable and domain-brittle. In ecology, the shift from raw species counts to diversity indices was preceded by demonstrations that raw counts were sampling-effort- dependent [2]. In both cases, the negative results were

not the end of the field—they were the beginning of a more productive phase.

The same pattern holds here. The module-graph representation, as currently implemented, is the "hand-crafted feature" of the yalumba program. It encodes biological intuition (k-mers that co-occur form functional modules) but lacks the engineering discipline (determinism, scale-invariance, measurement validation) needed for reliable science. The next phase replaces intuition-driven construction with measurement-validated construction.

> ▶ **Key Finding**
>
> **The most important finding in this paper is not that the representation is broken. It is that we now know exactly how it is broken, exactly which features are salvageable, and exactly what the acceptance criterion for a fixed representation must be.** The problem is now well-posed. That is progress.

The research thesis of the yalumba project—that human inheritance can be modeled as the transmission of resilient, interacting genomic ecosystems—is not invalidated by this audit. The thesis was never about module graphs or k-mer co-occurrence. It was about a level of biological organization that is richer than haplotype segments and deeper than token sharing. The module-graph representation was one attempt to access that level. It was not the right attempt, or at least not a sufficiently careful one. The audit tells us what "careful" means: deterministic, scale-invariant, and validated before use. The next representation that satisfies these criteria will be the foundation for a second decade of algorithmic development—and this time, the algorithms will be building on rock, not sand.

References

[1] Shai Ben-David, Ulrike von Luxburg, and Dávid Pál. A sober look at clustering stability. *Learning Theory (COLT)*, pages 5–19, 2006.

[2] J Roger Bray and J T Curtis. An ordination of the upland forest communities of southern Wisconsin. *Ecological Monographs*, 27(4):325–349, 1957.

[3] Karel Břinda et al. Genomic surveillance using reference-free phylogenetics. *Cell Genomics*, 3, 2023.

[4] Brian L Browning and Sharon R Browning. Improving the accuracy and efficiency of identity-by-descent detection in population data. *Genetics*, 194:459–471, 2013.

[5] Brian L Browning and Sharon R Browning. IBIS: identity by descent between sequential pairs of individuals. *American Journal of Human Genetics*, 108:1880–1890, 2021.

[6] Sharon R Browning and Brian L Browning. Identity by descent between distant relatives: detection and applications. *Annual Review of Genetics*, 46:617–633, 2012.

[7] Fan R K Chung. *Spectral Graph Theory*, volume 92 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, 1997.

[8] Herbert Edelsbrunner and John L Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010.

[9] Bradley Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

[10] Haoyu Fan et al. Reference-free and alignment-free approaches for estimating relatedness from whole-genome data. *Bioinformatics*, 37:2586–2593, 2021.

[11] Leland H Hartwell, John J Hopfield, Stanislas Leibler, and Andrew W Murray. From molecular to modular cell biology. *Nature*, 402:C47–C52, 1999.

[12] Christian Hennig. Cluster-wise assessment of cluster stability. *Computational Statistics & Data Analysis*, 52(1):258–271, 2007.

[13] Richard M Karp and Michael O Rabin. Efficient randomized pattern-matching algorithms. *IBM Journal of Research and Development*, 31(2):249–260, 1987.

[14] Peter Langfelder and Steve Horvath. WGCNA: an R package for weighted correlation network analysis. *BMC Bioinformatics*, 9:559, 2008.

[15] Facundo Mémoli. Gromov–Wasserstein distances and the metric approach to object matching. *Foundations of Computational Mathematics*, 11(4):417–487, 2011.

[16] Ron Milo, Shai Shen-Orr, Shalev Itzkovitz, Nadav Kashtan, Dmitri Chklovskii, and Uri Alon. Network motifs: simple building blocks of complex networks. *Science*, 298(5594):824–827, 2002.

[17] Masaaki Morisita. Measuring of interspecific association and similarity between communities. *Memoirs of the Faculty of Science, Kyushu University*, 3:65–80, 1959.

[18] Mikio Nakahara. *Geometry, Topology and Physics*. CRC Press, 2nd edition, 2003.

[19] Brian D Ondov et al. Mash: fast genome and metagenome distance estimation using MinHash. *Genome Biology*, 17, 2016.

[20] Gabriel Peyré and Marco Cuturi. *Computational Optimal Transport*, volume 11. Foundations and Trends in Machine Learning, 2019.

[21] Platinum Pedigree Consortium. CEPH 1463 pedigree datasets. `https://github.com/Platinum-Pedigree-Consortium/Platinum-Pedigree-Datasets`, 2024.